



Letter

A Genetic Algorithm-Based Layer Optimization Framework for Cross-Domain Fake News Detection Using RoBERTa

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Additional Information

EDICS

IF Information Forensics and Security / IF-FOR Multimedia forensics / IF-FOR-TEXT Text fingerprinting
 IF-FOR-TEXT Text fingerprinting

SL Speech and Language Processing / SL-REC Speech recognition / SL-REC-TEXT Text-based customization for speech-to-text

SL-REC-TEXT Text-based customization for speech-to-text

Subject Category

Machine learning and neural networks

Emerging Signal Processing Applications

Information Forensics and Security

If the manuscript has more than one author, briefly describe every author's significant intellectual contribution (max 25 words per author).

- Ms. Ishita Sharma: Conceived the study, supervised research activities, reviewed manuscript, and approved the final submission.
 - Dr. Abhishek Bhatt: Conducted experiments, collected data, and prepared the initial manuscript draft.
 - Vandana Thakur: Performed literature review, analyzed results, and contributed to manuscript editing.

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Explain in detail why the contribution of this manuscript is within the scope of IEEE Signal Processing Letters?

This manuscript is within the scope of IEEE Signal Processing Letters because it presents a novel signal and information processing framework for cross-domain fake news detection using transformer-based representation learning and evolutionary optimization. The proposed GA-based layer selection mechanism optimizes RoBERTa representations by identifying informative transformer layers and reducing redundant or domain-specific features, which relates directly to feature extraction, adaptive representation learning, and intelligent information processing. The work also addresses domain adaptation and negative transfer reduction, important topics in modern statistical signal processing and machine learning. Furthermore, the manuscript provides systematic experimental validation across multiple benchmark datasets and demonstrates significant performance improvements over existing baselines. The combination of transformer architectures, optimization techniques, and semantic signal analysis makes the contribution highly relevant to the emerging AI-driven research areas covered by IEEE Signal Processing Letters.

Why is the contribution significant (What impact will it have)?

The contribution is significant because it addresses the critical challenge of cross-domain fake news detection, where models trained on one dataset often fail to generalize to other domains. The proposed GA-based layer selection framework improves transformer efficiency by identifying informative RoBERTa layers while reducing redundant and domain-specific representations. This enhances model adaptability, minimizes negative transfer, and improves detection accuracy across heterogeneous datasets. The achieved performance gains demonstrate the framework's robustness for real-world misinformation detection scenarios such as social media monitoring, public health communication, and online news verification. Additionally, the layer-wise analysis provides interpretability into transformer behavior, contributing to explainable and efficient deep learning research. The proposed hybrid optimization approach can also be extended to other natural language processing and information processing tasks requiring domain adaptation and transferable semantic representation learning.

What are the three papers in the published literature most closely related to this paper?

- Entity-centric multi-domain transformer for improving generalization in fake news detection — Proposes a transformer-based framework for multi-domain fake news detection and domain generalization using domain-invariant and domain-specific representations.
 - MisRoBÆRTa: Transformers versus Misinformation — Introduces a RoBERTa-based transformer architecture for misinformation detection and demonstrates the effectiveness of transformer representations in fake news classification.
 - Performance Analysis of Transformer Based Models (BERT, ALBERT, and RoBERTa) in Fake News Detection — Provides comparative evaluation of transformer models for fake news detection and highlights the superior performance of RoBERTa-based approaches

What is distinctive/new about the current paper relative to these previously published works?

Unlike existing studies that primarily use fixed transformer architectures for fake news detection, the current paper introduces a novel Genetic Algorithm (GA)-based layer selection framework for RoBERTa to improve cross-domain generalization. Previous works mainly focus on transformer effectiveness within individual or limited domains, whereas this study specifically addresses negative transfer and domain adaptation across heterogeneous datasets. The proposed hybrid GA-transformer approach dynamically identifies informative layers and removes redundant or domain-specific representations, improving robustness and transferability. Additionally, the paper provides systematic cross-domain evaluation using ISOT as the source and COVID-19, GossipCop, and WELFake as target datasets, along with detailed layer-wise analysis to explain transformer behavior. The achieved performance improvements and interpretability distinguish this work from earlier transformer-based fake news detection approaches.

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

Name	Type of File	Size	Page
GA Based fake news detection.pdf	Main Document - PDF	1.6 MB	Page 5

A Genetic Algorithm-Based Layer Optimization Framework for Cross-Domain Fake News Detection Using RoBERTa

Ishita Sharma*, Dr. Abhishek Bhatt*, Vandana Thakur†

Abstract—Fake news detection has emerged as a critical challenge due to the rapid proliferation of misinformation on online platforms. While transformer-based models such as RoBERTa achieve strong performance in text classification, domain-adaptive training often leads to negative transfer when applied to heterogeneous target domains. To address this limitation, this paper proposes a Genetic Algorithm (GA)-based layer optimization framework for improving cross-domain robustness. The approach begins with domain-adaptive fine-tuning of RoBERTa on the ISOT dataset, followed by an evolutionary search strategy that selects an optimal subset of transformer layers to retain transferable knowledge while mitigating domain-specific bias. Unlike conventional fine-tuning, the proposed method adaptively filters redundant or over-specialized layers, enabling more effective generalization across domains. Experimental evaluations on COVID-19, GossipCop, and WELFake datasets demonstrate that the proposed framework consistently outperforms baseline models, achieving accuracies of 97.19%, 86.64%, and 99.00%, respectively. These results indicate that GA-based adaptive layer selection significantly reduces negative transfer and enhances cross-domain generalization with minimal reliance on target-domain supervision.

Index Terms—Fake news detection, RoBERTa, domain adaptation, negative transfer, genetic algorithm, layer optimization, transfer learning, cross-domain.

I. INTRODUCTION

THE rapid proliferation of misinformation across digital platforms has necessitated the development of robust automated fake news detection systems. Transformer-based architectures, particularly RoBERTa [1], have emerged as state-of-the-art solutions for various natural language processing tasks, including fake news classification. These models leverage large-scale pretraining on general corpora followed by task-specific fine-tuning, enabling superior contextual representation learning and classification performance.

Despite their success, a fundamental challenge in transfer learning arises when models pretrained on domain-specific data are applied to different but related domains, a phenomenon commonly referred to as negative transfer [2], [3]. Negative transfer occurs when knowledge learned from a source domain adversely affects performance in the target

domain, resulting in degradation compared to models initialized from general-purpose pretrained representations. Recent studies highlight that domain-adapted representations can introduce dataset-specific biases when the linguistic or topical distributions between the source and target domains differ significantly [3].

This issue is particularly critical in the detection of fake news due to the heterogeneous nature of misinformation across domains such as politics, healthcare, and entertainment. For example, COVID-19 misinformation datasets [4] and entertainment-oriented datasets such as GossipCop [5] exhibit substantially different writing styles, contextual cues, and deception patterns. Consequently, models pretrained exclusively on a single fake news dataset may overfit to source-domain characteristics, limiting their cross-domain generalization capability. Prior research has explored traditional machine learning and deep learning approaches [6]–[8]; however, effective transfer across heterogeneous datasets remains an open challenge.

Transformer architectures, introduced through the self-attention mechanism [9], enable models to capture long-range contextual dependencies in textual data. RoBERTa [1], a robustly optimized variant of BERT, improves representation learning through dynamic masking and large-scale training.

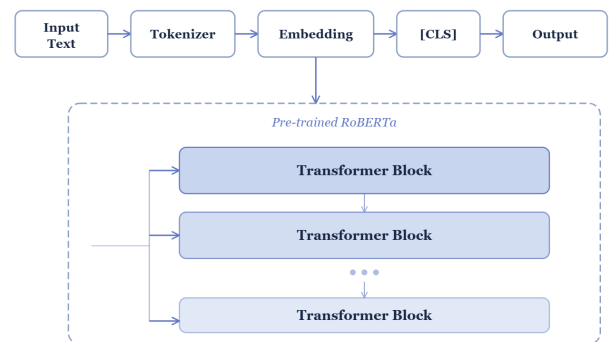


Fig. 1. RoBERTa architecture used for fake news classification.

As illustrated in Fig. 1, the input text is tokenized into subword units and passed through an embedding layer and multiple transformer blocks, where the final [CLS] token representation is used for classification. The RoBERTa-base model used in this work consists of 12 stacked transformer encoder layers, each with 12 attention heads and a hidden size of

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768. These layers learn hierarchical linguistic representations, where lower layers capture syntactic features and higher layers encode semantic and task-specific information.

In addition to transformer-based modeling, optimization techniques such as Genetic Algorithms (GA) are widely used for solving combinatorial optimization problems [10], [11]. These algorithms evolve a population of candidate solutions using operations such as selection, crossover, and mutation guided by a fitness function. As illustrated in Fig. 2 below, candidate solutions are iteratively refined across generations based on fitness evaluation, enabling efficient exploration of large search spaces.

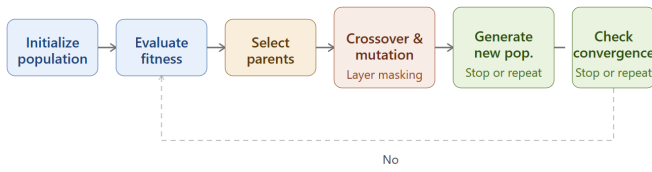


Fig. 2. Genetic Algorithm optimization process.

The layered structure of transformer models suggests that not all layers contribute equally to cross-domain adaptation. While lower layers capture general linguistic features, higher layers tend to encode domain-specific information. Therefore, uniformly fine-tuning all layers may lead to suboptimal performance in cross-domain settings.

Main Contributions: Based on the identified research gaps, the main contributions of this work are as follows:

- Proposes a GA-based layer selection framework for RoBERTa to improve cross-domain fake news detection.
- Develops a hybrid GA–transformer approach to filter redundant and domain-specific layers.
- Designs a systematic cross-domain evaluation using ISOT as the source and COVID-19, GossipCop, and WELFake as target datasets.
- Demonstrates superior performance over in-domain and cross-domain baselines, achieving accuracies of 97.19%, 86.64%, and 99.00%.
- Provides layer-wise analysis to explain the role of transformer layers in reducing negative transfer.

II. RELATED WORK

Fake news detection has been extensively studied using both traditional machine learning and deep learning approaches. Comprehensive surveys have highlighted key challenges in misinformation detection, including linguistic manipulation, contextual ambiguity, and rapidly evolving information ecosystems [6], [12]. Early deep learning approaches introduced hybrid neural architectures that combine textual and contextual information for misinformation detection, demonstrating improved performance over conventional feature-engineered methods [7], [13].

The emergence of transformer-based architectures significantly advanced natural language processing by enabling self-attention mechanisms capable of modeling long-range

contextual dependencies [9]. Building upon this framework, pretrained language models such as BERT [14], RoBERTa [1], XLNet [15], and ALBERT [16] achieved state-of-the-art performance across multiple NLP tasks. Transformer-based approaches have since been successfully applied to fake news detection, with models such as FakeBERT and domain-adapted transformers demonstrating substantial improvements over earlier neural architectures [8], [17].

Transfer learning has become a fundamental paradigm for adapting pretrained language models to downstream text classification tasks [2]. However, when the source and target domains differ significantly, negative transfer may occur, leading to degraded generalization performance [2], [3]. This issue is particularly critical in fake news detection, where datasets vary considerably in topic distribution, writing style, and semantic structure. Recent studies have explored domain adaptation techniques, including adversarial training and multi-domain learning, to improve cross-domain robustness [18], [19]. Further, multimodal fusion approaches that combine textual and visual signals have also been investigated to strengthen cross-domain fake news detection [20].

Recent research has focused on improving robustness and efficiency in transformer-based models. Parameter-efficient fine-tuning (PEFT) methods, such as adapters [21], LoRA [22], and prompt-based learning [23], aim to preserve pretrained knowledge while reducing computational overhead. Layer-wise fine-tuning strategies such as selective fine-tuning and progressive unfreezing have also been explored to balance knowledge retention and task adaptation [24]. In a related direction, automated layer freezing methods have been proposed to accelerate fine-tuning by identifying and fixing non-contributing model blocks, reducing unnecessary parameter updates [25]. Despite these advancements, most approaches rely on heuristic or manually selected configurations, limiting their adaptability across heterogeneous domains.

With the rapid evolution of large language models (LLMs), recent works have explored instruction-tuned and generative models for misinformation detection and reasoning tasks [26]–[30]. While these models demonstrate strong zero-shot and few-shot capabilities, they often require significant computational resources and may still suffer from domain-specific biases.

Evolutionary optimization techniques, particularly Genetic Algorithms, have demonstrated strong effectiveness in neural architecture search (NAS) and hyperparameter optimization problems involving large combinatorial search spaces [10], [31]. Their population-based search capability makes them well suited for optimizing structural components of deep neural networks. Recent studies in neural architecture search further highlight the effectiveness of evolutionary strategies in optimizing deep learning models [11].

Despite these advancements, limited research has explored automated layer-wise optimization of pretrained transformer models for mitigating negative transfer in fake news detection. Existing approaches typically employ full fine-tuning or predefined adaptation schemes that may not generalize effectively across heterogeneous domains. In contrast, this work proposes a Genetic Algorithm-based layer optimization framework that

automatically identifies the optimal subset of RoBERTa layers for robust cross-domain fake news detection.

III. DATASETS

This study proposes a Genetic Algorithm (GA)-based layer optimization framework for cross-domain fake news detection. The proposed method automatically determines the optimal subset of RoBERTa layers to fine-tune, enabling effective knowledge transfer while mitigating negative transfer across heterogeneous datasets. The framework is evaluated under a strict cross-domain protocol, where RoBERTa is initialized on the ISOT fake news dataset [32] and subsequently tested on three completely unseen target datasets: COVID-19, GossipCop, and WELFake.

The proposed framework is evaluated on these four publicly available fake news datasets covering diverse domains, writing styles, and content distributions. These datasets enable comprehensive evaluation of cross-domain generalization and robustness.

ISOT Fake News Dataset [32]: The ISOT dataset consists of news articles collected from both reliable and unreliable sources. It includes structured textual content with corresponding labels, where labels are represented in binary format (0=fake, 1=real). The dataset primarily contains political news articles and is used for domain-adapted fine-tuning to initialize the RoBERTa model. Its structured nature makes it suitable for learning domain-specific linguistic patterns.

COVID-19 Fake News Dataset [4]: This dataset comprises social media posts related to COVID-19 misinformation, reflecting real-world noisy and informal language. The original labels are converted into binary format for consistency. Due to differences in topic distribution, vocabulary, and writing style compared to ISOT, this dataset is used to evaluate cross-domain transfer robustness.

GossipCop Dataset [5]: The GossipCop dataset contains entertainment and celebrity news articles labeled as fake or real. It exhibits stylistic and contextual differences compared to political datasets, making it suitable for analyzing negative transfer effects in cross-domain settings.

WELFake Dataset [33]: WELFake is a large-scale dataset consisting of full-length news articles collected from diverse domains. Its size and heterogeneity make it suitable for large-scale evaluation and for assessing the generalization capability of the proposed framework.

IV. PROPOSED METHOD

The overall workflow of the proposed framework is illustrated in Fig. 3. The framework models the transformation of data from a source domain to multiple heterogeneous target domains, followed by optimization-based layer selection to improve cross-domain fake news detection.

As shown in Fig. 3 a pretrained RoBERTa-base model [1], consisting of 12 transformer encoder layers, is used as the base architecture for extracting contextual representations from textual data. The base model is further fine-tuned on the ISOT dataset [32], which serves as the source domain. This step enables the model to learn domain-specific linguistic and

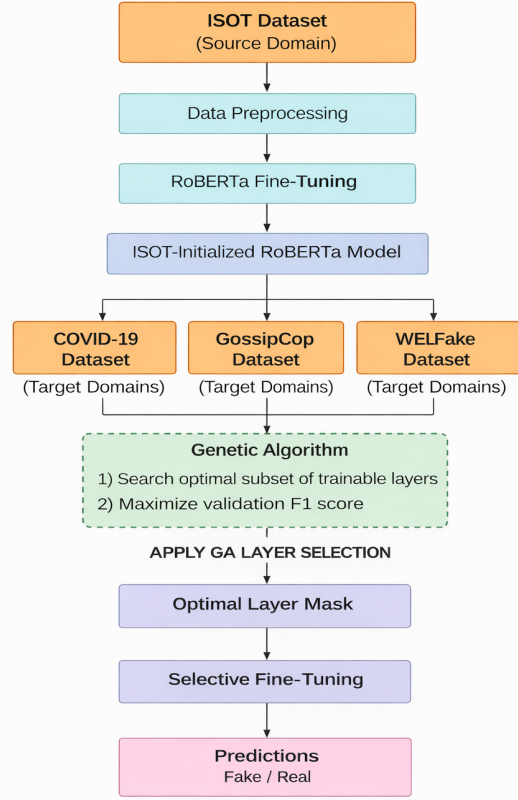


Fig. 3. Workflow illustrating dataset transformation from source domain (ISOT) to multiple target domains, followed by GA-based layer optimization for cross-domain fake news detection.

stylistic patterns associated with fake and real news, resulting in an ISOT-initialized RoBERTa model. During Cross-Domain Transfer, the ISOT-initialized model is transferred to target datasets, including COVID-19, GossipCop, and WELFake. Due to variations in topic distribution, writing style, and contextual structure, direct fine-tuning may lead to negative transfer [2], [3], causing performance degradation.

Further to implement Genetic Algorithm-Based Layer Optimization, a Genetic Algorithm (GA) [10], [31] is employed to identify the optimal subset of transformer layers for fine-tuning. Each candidate solution is represented as a binary mask over layers, and the objective is to maximize the validation F1-score. Genetic Algorithms are well suited for such combinatorial optimization problems due to their population-based search strategy [11]. Finally the optimal layer configuration obtained from the GA is used to fine-tune the model on the target training dataset. The final model is then evaluated on the corresponding test dataset to assess performance.

A. Dataset Preprocessing

All datasets are processed using a unified preprocessing pipeline to ensure consistency across domains, as illustrated in Fig. 4. Initially, samples with missing text or labels are removed, followed by duplicate elimination to avoid data redundancy. The labels from all datasets are converted into a binary format (fake = 0, real = 1) for uniform classification.

The textual data is then tokenized using the RoBERTa tokenizer, which converts raw text into subword tokens suitable for transformer-based models. To maintain fixed input dimensions, sequences are padded or truncated to a maximum length of 256 tokens. Finally, the processed data is converted into PyTorch tensor format for efficient model training.

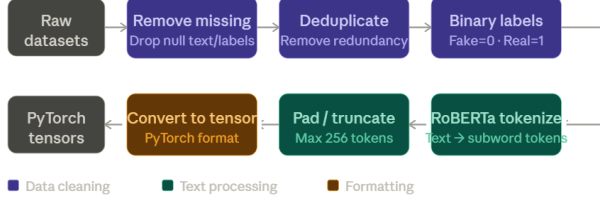


Fig. 4. Dataset preprocessing pipeline including data cleaning, tokenization, padding/truncation (256 tokens), and tensor conversion.

B. Model Architectures

1) *Vanilla RoBERTa Baseline*: Let M_θ denote the RoBERTa-base model parameterized by θ , consisting of $L = 12$ transformer encoder layers and a task-specific classification head. Given an input text sequence $X = \{x_1, x_2, \dots, x_n\}$, contextual representations are generated through stacked self-attention layers.

The hidden representation corresponding to the classification token h_{cls} is passed through a linear classifier:

$$\hat{y} = \text{Softmax}(W h_{cls} + b) \quad (1)$$

where $W \in \mathbb{R}^{d \times C}$ and $b \in \mathbb{R}^C$ are learnable parameters.

In the baseline configuration, the top four transformer layers and the classification head are fine-tuned directly on each target dataset:

$$D_T \in \{D_{\text{COVID-19}}, D_{\text{GossipCop}}, D_{\text{WELFake}}\}$$

This in-domain fine-tuning achieves accuracies of 96.40%, 86.31%, and 98.73% on COVID-19, GossipCop, and WELFake respectively, and serves as the upper-bound reference baseline.

Binary cross-entropy loss is used:

$$\mathcal{L} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (2)$$

2) *ISOT-Initialized RoBERTa and Negative Transfer*: To incorporate fake news domain knowledge, RoBERTa is first fine-tuned on the ISOT dataset [32]:

$$\theta_{\text{ISOT}} = \text{FineTune}(\theta_{\text{base}}, D_{\text{ISOT}}) \quad (3)$$

This stage enables the model to learn stylistic and semantic patterns characteristic of fake news.

However, when transferring θ_{ISOT} to new datasets, negative transfer may occur. We empirically observe significant performance degradation when adapting to $D_{\text{COVID-19}}$ and $D_{\text{GossipCop}}$, with accuracy dropping to 63.50% and 76.06% respectively, compared to the in-domain Vanilla baseline of 96.40% and 86.31%. These datasets differ significantly from ISOT in topic distribution and writing style. In contrast, WELFake

Algorithm 1 GA-Based Layer Mask Optimization

Require: $\theta_{\text{ISOT}}, D_T, N, G$

Ensure: Optimal mask m^*

- 1: Initialize population of N random masks
- 2: **for** $g = 1$ **to** G **do**
- 3: **for** each chromosome c **do**
- 4: Apply mask c to θ_{ISOT}
- 5: Fine-tune selected layers on D_T for one epoch
- 6: Compute validation F1-score $F(c)$
- 7: **end for**
- 8: Perform selection, crossover, mutation, and elitism
- 9: **end for**
- 10: **return** chromosome with highest fitness

exhibits greater similarity to ISOT, and while a moderate performance gap is still observed (95.11% vs 98.73%), the degradation is considerably less severe compared to COVID-19 and GossipCop.

These findings motivate selective layer adaptation rather than uniform fine-tuning.

3) *Transfer Learning and Layer-Wise Optimization*: The domain-adapted model θ_{ISOT} is transferred to a target dataset D_T . Instead of fine-tuning all transformer layers, we perform selective fine-tuning.

Let $\theta_{\text{ISOT}}^{(l)}$ denote the parameters of the l -th transformer layer, where $l \in \{1, 2, \dots, L\}$.

We define a binary mask:

$$m \in \{0, 1\}^L \quad (4)$$

where $m_l = 1$ indicates layer l is trainable, and $m_l = 0$ indicates layer l is frozen.

The set of trainable parameters under mask m is:

$$\theta_{\text{train}}(m) = \{\theta_{\text{ISOT}}^{(l)} \mid m_l = 1\} \quad (5)$$

The objective is to determine the optimal mask:

$$m^* = \arg \max_{m \in \{0, 1\}^L} F1_{\text{val}}(M_{\theta_{\text{train}}(m)}) \quad (6)$$

Since the search space contains 2^L possible layer configurations ($2^{12} = 4096$ for RoBERTa-base), exhaustive search is computationally infeasible.

4) *Genetic Algorithm Formulation*: To efficiently explore this space, a Genetic Algorithm [10], [11] is used. Each chromosome represents a layer mask:

$$c \in \{0, 1\}^L \quad (7)$$

Fitness is defined as the validation F1-score:

$$F(c) = F1_{\text{val}}(M_{\theta_{\text{train}}(c)}) \quad (8)$$

The GA evolves using selection, crossover, mutation, and elitism to find the optimal configuration:

$$m^* = \arg \max_c F(c) \quad (9)$$

Once the optimal mask m^* is identified, the ISOT-initialized model is fine-tuned using this configuration on the full training dataset, followed by evaluation on the test dataset.

Algorithm 2 Final Training Using Optimal Mask**Require:** $\theta_{\text{ISOT}}, m^*, D_T^{\text{train}}, D_T^{\text{test}}$

- 1: Apply mask m^* to θ_{ISOT}
- 2: Fine-tune model on D_T^{train}
- 3: Evaluate on D_T^{test}

The complexity of the proposed method is given by:

$$\mathcal{O}(G \times N \times (T_{\text{train}} + T_{\text{eval}})) \quad (10)$$

where G is the number of generations and N is the population size.

C. Model Training and evaluation

Each dataset is divided into training, validation, and test sets to ensure unbiased evaluation. Specifically, 80% of the data is used for training, while the remaining 20% is split equally into validation and test sets (10% each). Stratified sampling is employed to preserve class distribution across all splits.

The validation set is used during the proposed GA framework for evaluating candidate layer configurations, while final performance is reported on the unseen test dataset.

Training Strategy The detailed training configuration is summarized in Table I.

TABLE I
UNIFIED TRAINING CONFIGURATION

Parameter	ISOT	GA Eval.	Final
Epochs	3	1	4
Learning rate	2×10^{-5}	2×10^{-5}	2×10^{-5}
Train batch size	16	16	16
Eval batch size	16	16	16
Evaluation strategy	Epoch	Epoch	Epoch
Checkpoint saving	Yes	No	Yes
Mixed precision	FP16	Disabled	FP16
Optimizer	AdamW	AdamW	AdamW
Load best model	No	No	Yes

The training process is performed in three sequential stages, as illustrated in Fig. 5.

Stage 1: RoBERTa is fine-tuned on the ISOT dataset to obtain domain-adapted parameters (θ_{ISOT}).

Stage 2: A Genetic Algorithm evaluates different transformer layer subsets via partial fine-tuning using validation F1-score as the fitness function.

Stage 3: The optimal layer configuration is fully trained on target datasets and evaluated on unseen test data to assess generalization performance.

1) *Evaluation Metrics:* Model performance is evaluated using standard classification metrics. Accuracy measures overall correctness, while Precision and Recall evaluate prediction quality for positive samples. The F1-score is used as the primary optimization metric.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

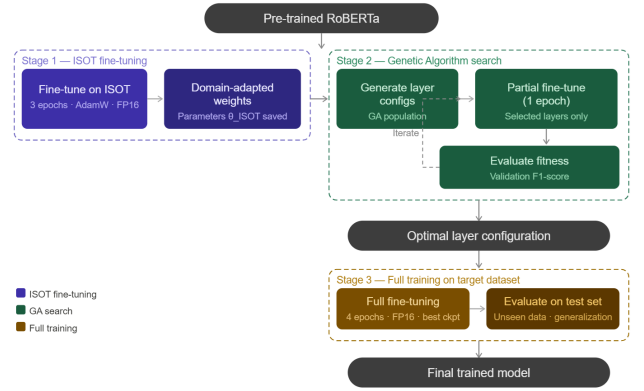


Fig. 5. Training strategy consisting of ISOT fine-tuning, GA-based layer optimization, and final training using the optimal layer configuration.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

The validation F1-score is used during the GA optimization stage, while final results are reported on the test dataset.

D. Proposed Framework using Genetic Algorithm Configuration

The proposed Genetic Algorithm (GA) based layer optimization process for selecting the optimal subset of transformer layers using validation F1-score as the fitness function is illustrated in Fig. 6 and the configuration is summarized in Table II. Each chromosome is represented as a binary vector

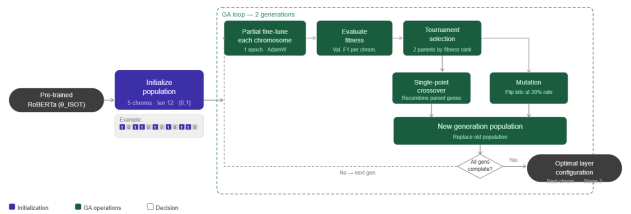


Fig. 6. GA framework used for layer optimization.

TABLE II
GENETIC ALGORITHM PARAMETERS

Parameter	Value
Chromosome length	12
Population size	5
Generations	2
Parents per mating	2
Gene space	{0, 1}
Mutation rate	20%
Selection method	Tournament
Crossover	Single point
Fitness function	Validation F1 score

of length 12, corresponding to the transformer layers, where 1 indicates a trainable layer and 0 indicates a frozen layer. The algorithm initializes a population of candidate solutions and iteratively improves them using selection, crossover, and mutation operations. During each iteration, partial fine-tuning is performed for one epoch, and the validation F1-score is used as the fitness function. The best-performing chromosome is selected as the optimal layer configuration.

V. RESULTS AND DISCUSSION

This section evaluates the effectiveness of the proposed GA-based framework for cross-domain fake news detection. The analysis focuses on negative transfer mitigation, optimization behavior, and performance improvements achieved through adaptive transformer layer selection. The proposed framework effectively mitigates the limitations of conventional transfer learning, where partial fine-tuning of the ISOT-initialized RoBERTa model often leads to performance degradation under domain shift. Such degradation, particularly evident in datasets like COVID-19 and GossipCop, arises from biases in topic distribution and linguistic style, resulting in reduced Macro F1-scores and imbalanced class predictions.

In contrast, the proposed approach employs a Genetic Algorithm for adaptive layer selection, retaining only those transformer layers that contribute positively to target-domain generalization. This strategy suppresses source-domain bias and enables more effective knowledge transfer across heterogeneous datasets. Consequently, the model achieves improved class balance and overall predictive performance, demonstrating its robustness and effectiveness in cross-domain fake news detection scenarios. The convergence behavior of the proposed GA framework is illustrated in Fig. 7. The validation F1-score improves progressively across generations, indicating effective exploration of the transformer layer configuration space.

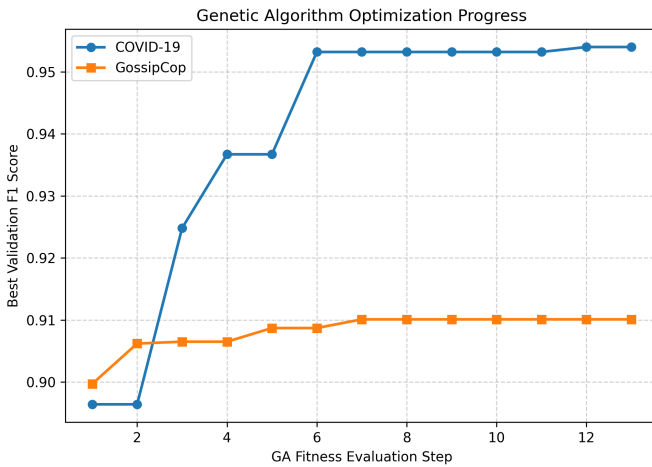


Fig. 7. Convergence of the proposed GA framework showing validation F1-score improvement across generations.

Despite the combinatorial nature of the search space, the framework achieves high-quality configurations within the constrained two-generation budget. This demonstrates that the proposed optimization strategy remains computationally

feasible under constrained search settings while still capable of identifying high-performing layer subsets.

The proposed method successfully restores performance by selectively fine-tuning relevant transformer layers. For the COVID-19 dataset, the performance of the proposed method closely matches that of the vanilla RoBERTa model. In the GossipCop dataset, the proposed method surpasses both baseline approaches, indicating improved robustness in handling stylistic variations.

The use of validation F1-score as the fitness function ensures that the selected configurations are directly aligned with classification performance.

Fig. 8 performance across datasets using the Weighted F1-score. It has been observed that the Proposed (RoBERTa + GA) method consistently outperforms the baseline ISOT Init across all datasets. There is a significant improvement in the GossipCop dataset, where the Weighted F1 score jumps from 66.0% (Baseline) to 91.67% (Proposed). On the COVID-19 and WELFake datasets, the Proposed method achieves Weighted F1 scores (97.35% and 99.65% respectively) that are even higher than the Vanilla model's in-domain performance, showcasing exceptional generalization capabilities. presents a comparison of model

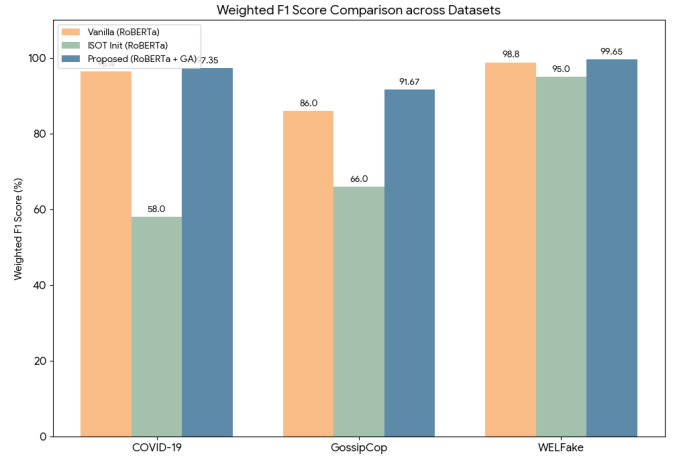


Fig. 8. Performance comparison of models highlighting improvements achieved by the proposed method.

As given in Table III Comparison across training strategies highlights the limitations of fixed transfer learning approaches. Vanilla RoBERTa is fine-tuned directly on each target dataset as an *in-domain* baseline, providing an upper-bound reference for each dataset. In contrast, ISOT initialization follows a *cross-domain* protocol, where the model is pre-trained on the ISOT source domain before being adapted to the target datasets, introducing domain-specific bias that may not generalize effectively across heterogeneous datasets.

The proposed GA framework dynamically determines the optimal subset of transformer layers for adaptation, balancing knowledge preservation and domain specialization. This results in improved generalization across datasets with varying linguistic characteristics.

For the WELFake dataset, which shares similarities with the ISOT dataset, all models achieve high performance. How-

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS**

Model	Dataset	Eval Type	Acc (%)	Prec (%)	Rec (%)	Macro F1	Weighted F1
<i>Existing Methods (Published, In-Domain)</i>							
CT-BERT + BiGRU [34]	COVID-19	In-domain	98.46	97.97	99.11	98.54	—
CNN [35]	COVID-19	In-domain	94.20	—	—	—	—
RoBERTa + XGBoost [36]	GossipCop	In-domain	99.30	98.94	99.16	99.05	99.30
BERT + LSTM [8]	GossipCop	In-domain	84.10	89.00	91.00	89.00	—
WELFake [33]	WELFake	In-domain	96.73	94.60	98.61	96.56	—
FakeBERT [8]	WELFake	In-domain	98.90	—	—	—	—
<i>Ablation Baselines (Our Experiments)</i>							
Vanilla (RoBERTa) [1]	COVID-19	In-domain	96.40	96.20	96.60	96.30	96.40
ISOT Init (RoBERTa) [32]	COVID-19	Cross-domain	63.50	75.00	64.00	57.00	58.00
Vanilla (RoBERTa) [1]	GossipCop	In-domain	86.31	82.00	78.00	80.00	86.00
ISOT Init (RoBERTa) [32]	GossipCop	Cross-domain	76.06	38.00	50.00	43.00	66.00
Vanilla (RoBERTa) [1]	WELFake	In-domain	98.73	98.60	98.90	98.70	98.80
ISOT Init (RoBERTa) [32]	WELFake	Cross-domain	95.11	95.00	95.00	95.00	95.00
<i>Proposed Method (Cross-Domain)</i>							
Proposed (RoBERTa + GA)	COVID-19	Cross-domain	97.19	96.17	98.57	97.00	97.35
Proposed (RoBERTa + GA)	GossipCop	Cross-domain	86.64	87.23	96.58	79.00	91.67
Proposed (RoBERTa + GA)	WELFake	Cross-domain	99.00	99.60	99.70	99.60	99.65

— indicates that the metric was not reported in the original publication.

**Existing methods and Vanilla RoBERTa are evaluated under an in-domain setting, whereas ISOT Init and the proposed method follow a cross-domain protocol with initialization on ISOT and evaluation on unseen target datasets.

ever, the proposed method maintains competitive performance while avoiding unnecessary parameter updates, demonstrating efficient adaptation.

The optimal transformer layer configurations identified by the proposed GA framework as illustrated in Fig. 9 demonstrate that different datasets require different combinations of transformer layers for optimal performance. Unlike fixed layer selection strategies, the proposed approach selects layers across both lower and higher levels of the transformer architecture. This indicates that both general linguistic features and domain-specific representations are important for effective cross-domain adaptation.

Furthermore, Table III shows that existing methods are evaluated under an *in-domain* setting, where training and testing data share the same distribution [8], [33]–[36]. In contrast, ISOT Init and proposed framework follow a *cross-domain* protocol, where the model is trained on the ISOT dataset and evaluated on completely unseen target datasets. This makes the task significantly more challenging due to domain shifts in content, style, and vocabulary. Despite this, the proposed method achieves competitive or superior performance, demonstrating strong generalization capability.

Vanilla RoBERTa achieves strong performance in the in-domain setting, serving as an upper-bound reference. However, ISOT initialization significantly degrades performance under cross-domain conditions, with accuracy dropping to 63.50% on COVID-19, 76.06% on GossipCop, and 95.11% on WELFake, confirming the impact of negative transfer.

The proposed GA-based framework outperforms both base-

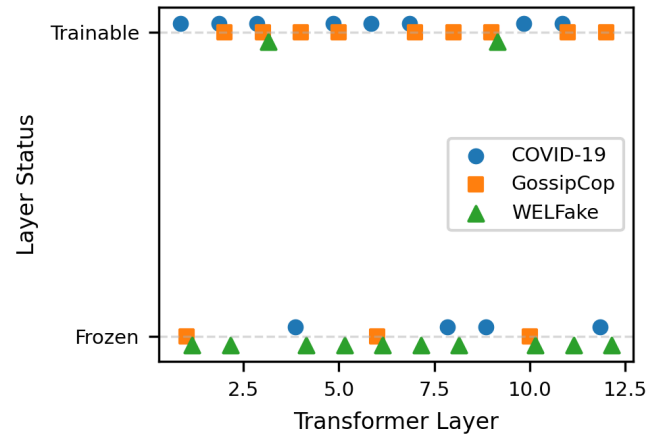


Fig. 9. Optimal transformer layer configurations selected by the proposed GA framework.

lines across all datasets. It achieves 97.19% accuracy on COVID-19, surpassing the Vanilla baseline (96.40%) despite operating in a cross-domain setting. On GossipCop, it improves performance to 86.64% accuracy and a weighted F1-score of 91.67%, with high recall (96.58%). On WELFake, it achieves 99.00% accuracy and a weighted F1-score of 99.65%, outperforming the Vanilla baseline (98.73%).

Overall, the results demonstrate that the proposed method effectively mitigates negative transfer and achieves superior cross-domain generalization compared to both in-domain and

cross-domain baselines.

VI. CONCLUSION

This paper presents a Genetic Algorithm (GA)-based layer selection framework for enhancing cross-domain fake news detection using domain-adapted transformer models. The results indicate that conventional partial fine-tuning of an ISOT-initialized RoBERTa model is sensitive to domain shift and can lead to negative transfer, particularly for datasets with significant differences in topic distribution and linguistic style.

The proposed approach mitigates this issue by adaptively selecting a subset of transformer layers for target-domain adaptation, thereby reducing source-domain bias while preserving transferable representations. This selective strategy results in more stable and effective cross-domain performance compared to uniform fine-tuning. Experimental results demonstrate consistent improvements over both in-domain and cross-domain baselines, achieving accuracies of **97.19%**, **86.64%**, and **99.00%** on the COVID-19, GossipCop, and WELFake datasets, respectively.

Despite these improvements, the framework is constrained by the computational cost of the GA search and its focus on textual features. Future work will explore more efficient optimization strategies, multimodal extensions, and evaluation in dynamic real-world environments to further enhance robustness and applicability.

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